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Location : New york

Sr. Big Data Engineer

PROFESSIONAL SUMMARY:

- Designed and deployed production-grade agentic retrieval systems using **Python** and **DSPy** to build intelligent agents that automatically pull PR metadata, commit history, and diffs from **Git** repositories, integrating LLM-driven evaluation frameworks while implementing prompt chaining and context packing techniques for accurate CES metric computation.
- Built and maintained robust data orchestration pipelines using **Dagster** and **Apache Airflow** on **AWS**, leveraging **ECS**, **EventBridge**, **Lambda**, and **S3** to ingest PR metadata, commit records, and branch tracking information, implementing backfill procedures, retries, and monitoring systems to ensure high reliability and data quality across engineering workflows.
- Implemented advanced LLM prompt optimization frameworks using **DSPy** to develop eval set creation, execution, and feedback-driven correction systems, integrating automated score correction mechanisms and structured dataset preparation pipelines while managing LLM call performance through batching strategies and caching layers for cost-efficient CES scoring.
- Developed data modeling architectures for PR and commit linking systems using **Python** and **AWS Glue**, establishing epic-level associations, branch merges, and work-type classification, implementing anomaly detection algorithms to identify data quality issues while ensuring backend services compute CES metrics accurately across distributed engineering teams.
- Created backend integration services using **Python**, **AWS Glue**, **Lambda**, and **S3** to ingest PR and commit metadata, compute LLM-driven code quality metrics, and expose results via APIs and UI endpoints, implementing observability layers with metrics, traces, and logs to maintain real-time pipeline health visibility.
- Engineered eval and feedback loop systems with **DSPy** frameworks, designing evaluation set structures, automated score correction workflows, and training data pipelines that transform feedback into structured datasets, integrating backend services to close the loop between PR-level scoring, context retrieval, and CES metric computation.
- Optimized LLM prompting and context retrieval using **DSPy**, implementing prompt chaining architectures and context packing algorithms for Git diffs, commit history, and PR metadata, measuring prompt effectiveness through cost and quality tradeoff analysis, and integrating agentic retrieval agents to support CES Phase 3 scoring improvements.
- Built scalable AWS-native data pipelines using **ECS**, **EventBridge**, **Lambda**, **S3**, **Glue**, and **Dagster** to handle PR ingestion, commit tracking, and branch merge monitoring, implementing robust backfill procedures, retries, and high-throughput metadata processing to support distributed engineering teams at scale.
- Designed metadata classification systems using **Python**, **Jira**, and **Git**, assigning role labels and work-type categories to PRs and commits, developing filter and explorer features to query CES metrics, and implementing graph-based linking algorithms to improve epic-level tracking and engineering productivity visibility.
- Established observability and monitoring frameworks using **AWS CloudWatch** and **Dagster**, implementing metrics collection, distributed tracing, centralized logging, and data quality alerting systems to detect anomalies in PR ingestion, commit tracking, and CES metric computation, providing real-time dashboards for operational excellence.
- Integrated **SonarQube** with backend services to enrich PR scoring models, ingesting static analysis metrics, Git metadata, and commit history via **Dagster** pipelines on **AWS**, implementing classification algorithms to assess technical debt and quality degradation patterns for comprehensive engineering insights.
- Optimized LLM call performance using **DSPy** frameworks, implementing batching strategies, response caching, and evaluation frameworks to reduce operational cost, developing backend services for intelligent request routing between LLM providers, while monitoring token usage, latency, and prediction quality across large-scale PR pipelines.
- Developed graph-based artifact linking systems using **Python**, **Jira**, and **GitHub**, establishing relationships across epics, PRs, commits, and branches, implementing backend services to compute aggregated CES metrics and quality scores, supporting complex queries across linked artifacts for engineering productivity and code quality analysis.
- Implemented historical context retrieval agents with **DSPy**, **Python**, and **Git**, automatically extracting code evolution history, past PR discussions, and commit patterns to enhance LLM-driven scoring, building backend services and pipelines that prepare context for prompt engineering and improve CES metric accuracy.

- Created training data preparation workflows using **Dagster**, **Python**, and **S3**, transforming user feedback, manual score corrections, and eval set results into structured datasets for LLM fine-tuning, maintaining versioned datasets while tracking feedback provenance and quality labels for continuous improvement of agentic retrieval systems.
- Built end-to-end data pipelines in **Dagster**, orchestrating PR ingestion, commit linking, and branch-to-main tracking, implementing backfill strategies, retry logic, and monitoring dashboards, integrating data quality alerts for schema violations and missing commit linkages, while migrating legacy **Apache Airflow** DAGs for improved orchestration efficiency.
- Developed DSPy-based prompt optimization workflows to tune LLM strategies for CES scoring, creating eval sets from production PRs, running iterative prompt tuning experiments, integrating signature-based prompt chaining and context packing, measuring effectiveness, and closing feedback loops to improve scoring accuracy while reducing inference costs.
- Engineered observability infrastructure using **Python**, **Dagster**, and **AWS Lambda**, emitting metrics, traces, and logs to centralized platforms, building data quality alerts for schema drift and pipeline failures, and collaborating with SRE teams to establish runbooks, ensuring rapid incident response and sustained reliability across production-grade pipelines.
- Enhanced metadata classification and explorer features using **Python**, **Jira**, and **Git**, developing role and work-type models, applying rule-based and ML-driven logic to tag PRs, integrating graph-based linking for exploration, and backfilling historical PR and commit datasets, enabling CES scoring, quality metrics, and productivity dashboards across engineering teams.

TECHNICAL SKILLS:

- **Databases & Tools:** Teradata 15/14, MySQL, Oracle 11g/10g, MS SQL Server 2014/2012, Erwin 8/9
- **Cloud Environment:** GCP (Cloud storage, Bigquery, DataProc, GKE, Spanner, Functions), AWS (S3, Glue, EMR, Redshift, Lambda, Athena, EC2, Snowflake, RDS, Cloud Watch, VPC, SQS, Jenkins), Azure (Data factory, Databricks, SQL, Data lake, Blob storage, HDInsight, CosmosDB, Fabric)
- **Data Warehousing & BI:** Splunk, Star and Snowflake schema, Cloudera, Facts and Dimensions tables, SSIS
- **Big Data Ecosystem:** Kafka, Spark, Airflow, Hive, HDFS, NiFi, Spark Streaming, Zookeeper, Sqoop, Flume, MapReduce, Yarn
- **Reporting Tools:** Tableau, Semarchy, PowerBI, EHR
- **ETL:** AWS Glue, Informatica Power Center 10.x/9.6, Semarchy xDM, Talend 5.6, SSIS
- **Software Life Cycle/ Methodologies:** Agile models, Waterfall, SDLC
- **Python Libraries / Machine Learning Algorithms:** NumPy, Pandas, Scikit-Learn, Matplotlib, Gradient Boosting, Random Forest, Logistic Regression, Decision Tree
- **Containerization and Orchestration Tools:** Kubernetes, Docker Hub, Jenkins, Docker
- **Programming languages:** Scala, Python, Java, Pyspark, SQL, Spring boot, T-SQL, Unix
- **Certifications:** Tableau certified, Azure DP-203, ServiceNow CAD and SaaS Analyst, Azure AZ-900

PROFESSIONAL EXPERIENCE:

Humana: August 2023 – Current (Location: Louisville, KY)

Role: Sr. Big Data Engineer

Responsibilities:

- Designed and deployed agentic retrieval workflows using **Python**, **LangChain**, and **LangGraph** to build context-aware agents ingesting PR metadata, commit metadata, and **GitHub** history, enabling automated CES scoring and quality metric computation for healthcare pipelines while ensuring HIPAA-compliant audit trails and HL7 interoperability across **AWS S3**, **AWS Lambda**, and **AWS Glue**.
- Migrated legacy orchestration from **Apache Airflow** to **Dagster** across 40+ production pipelines processing healthcare claims, clinical records, and patient data on **AWS ECS** and **AWS EKS**, implementing retries, backfills, scheduling, and data quality alerts while exposing metrics, traces, and logs through **Power BI** and **Collibra** frameworks.
- Built production-grade LLM-driven scoring systems using **DSPy**, integrating prompt optimization, chaining, and context packing to automate healthcare PR-level scoring and quality metrics, leveraging **GPT-4**, **Claude**, **AWS Sagemaker**, and **MLflow** for model versioning, eval set generation, and feedback loops under HIPAA-compliant governance.

- Engineered backend services using **Python**, **AWS Lambda**, **AWS EventBridge**, and **AWS CDK** to ingest PR and commit metadata from **GitHub** and **Jenkins**, linking artifacts to compute CES metrics, quality signals, and anomalies while exposing results through RESTful APIs and **Power BI** dashboards under HIPAA compliance.
- Developed data modeling frameworks using **Pandas**, **Numpy**, and **MLLIB** to link PRs, commits, branches, and epics for healthcare software delivery metrics, building graph-based relationships across **Jira**, **GitHub**, and **AWS S3**, implementing anomaly detection, role classification, and effort estimation for HIPAA-compliant reporting.
- Orchestrated large-scale backfills and historical corrections using **Dagster** and **AWS Glue**, processing over 10 million commits and clinical record deltas with batching, caching, and DSPy-based score optimization, ensuring data quality alerts, metrics, traces, and logs captured in **AWS CloudWatch** and **Collibra** lineage tracking.
- Implemented **SonarQube** integration for code-quality scoring, linking static analysis results with **GitHub** PR metadata and commit history to enrich CES metrics, while building **Jira**-integrated workflows for epic/work-type classification and automated tagging, ensuring HIPAA-compliant metadata governance across **AWS ECS**, **AWS Lambda**, and **AWS S3**.
- Designed eval and feedback loops using **Python**, **DSPy**, and **MLflow** for LLM-driven prompt optimization, creating versioned evaluation datasets, collecting feedback from healthcare engineers via **Power BI** and **Jira**, closing loops to improve CES scoring accuracy and model explainability under HIPAA and HITECH mandates.
- Built observability and monitoring solutions using **AWS CloudWatch**, **AWS Lambda**, and **Dagster** asset sensors to track metrics, traces, logs, and data quality alerts across healthcare ETL pipelines, enabling automated alerting for failures, anomaly detection, SLA breaches, and ensuring regulatory audit readiness with **Collibra** and **ERWIN** lineage tracking.
- Developed agentic coding frameworks using **LangChain**, **LangGraph**, and **DSPy** to build retrieval agents that processed Git diffs, code context, and commit metadata, integrating with **AWS S3**, **AWS Glue**, and **AWS Lambda** to improve CES scoring, reduce LLM latency, and maintain HIPAA-compliant healthcare data processing.
- Constructed end-to-end data pipelines using **Dagster**, **AWS Glue**, **AWS S3**, and **AWS RDS** to ingest, transform, and load healthcare claims, clinical records, and patient data from HL7 FHIR sources, implementing retries, scheduling, backfills, and data quality alerts with **AWS EventBridge** triggers and HIPAA-compliant logging.
- Optimized LLM performance using batching, caching, and **DSPy**-based prompt optimization for healthcare code quality scoring, leveraging **AWS Sagemaker** and **MLflow** to tune prompt chaining, context packing, and eval execution workflows, balancing cost/quality tradeoffs while maintaining HIPAA-compliant model governance and HL7 interoperability.
- Integrated **GitHub**, **Jenkins**, **Jira**, and **SonarQube** metadata into unified data models using **Python**, **Pandas**, **AWS Glue**, and **ERWIN** for entity-relationship design, building graph-based linking across PRs, commits, branches, and epics to support role/work-type classification, epic linking, anomaly detection, and HIPAA-compliant analytics exposure via **Power BI**.
- Deployed **AWS CDK** infrastructure for pipelines and backend services on **AWS EC2**, **AWS EKS**, **AWS Lambda**, and **AWS S3**, automating provisioning, monitoring, and scaling of Dagster orchestrators, LLM inference endpoints, and **AWS Glue** jobs with metrics, traces, logs, and data quality alerts ensuring HIPAA and HITECH compliance.
- Collaborated with healthcare engineering teams to define CES metrics, quality scoring models, and eval sets using **Python**, **Dagster**, **AWS Lambda**, **AWS S3**, **Power BI**, and **Collibra**, translating PR metadata ingestion, commit tracking, and anomaly detection requirements into compliant backend integration, analytics, and data modeling solutions aligned with HIPAA, HITECH, and HL7 standards.
- Developed and deployed AI/ML pipelines using **MLLIB** and **Numpy** to predict patient readmission risk, orchestrated with **Airflow** for ingestion, transformation, and model retraining, ensuring HIPAA-compliant data handling, HL7 interoperability, and audit trails in alignment with HITECH Act regulations for healthcare systems.
- Engineered AI/ML clinical decision support using LLMs and **MLLIB**, processing unstructured EHR data, identifying high-risk patients, and deploying inference infrastructure on **AWS EC2** with encryption, role-based access, and logging to meet HIPAA and HITECH security requirements for real-time healthcare workflows.
- Designed and optimized relational schemas using **E/R Studio** for patient demographics, encounters, and lab results, ensuring normalization and HL7 FHIR compliance, generating data dictionaries and impact analysis reports for HIPAA audit readiness, enabling collaboration across clinical informatics teams while maintaining secure data exchange practices.
- Built predictive models for disease progression using **MLLIB** and **Numpy**, processing large patient datasets for feature engineering, statistical analysis, and evidence-based treatment plans, implementing de-identification, secure access policies, and documentation to meet HIPAA and HITECH standards while supporting clinical quality improvement initiatives.
- Implemented conversational AI chatbots using **LLMs** and **AWS EC2** to automate patient triage and scheduling, integrating with EHR systems, ensuring HIPAA privacy and HITECH compliance through secure API gateways, encryption, and access logging while enabling real-time healthcare interaction workflows across multi-site hospital networks.
- Migrated legacy workloads to **AWS EC2**, configuring VPCs, security groups, and identity policies to meet HIPAA technical safeguards, encrypt patient data, enforce access controls, maintain audit logs, implement VPN tunnels, multi-factor

authentication, and intrusion detection systems while ensuring HITECH breach notification and distributed healthcare compliance.

- Standardized hospital analytics data modeling using **E/R Studio**, creating entity-relationship diagrams, documenting normalization rules and referential constraints to support HL7 message exchanges, HIPAA compliance audits, onboarding, and future enhancements for population health management and quality reporting across healthcare IT systems.
- Performed numerical computation and statistical validation using **Numpy** on clinical trial data, supporting biostatisticians in patient outcome analysis, treatment efficacy, and adverse event reporting, ensuring HIPAA compliance, documenting methods, validating outputs, and aligning processes with FDA, HITECH, and institutional review board regulations for protected health information.

Environment: AWS Sagemaker, AWS Lambda, Dagster, Jira, Collibra, Numpy, ERWIN, Pandas, MLflow, MLLIB, AWS EC2, AWS RDS, S3, Lambda, DSPy, Jenkins, EventBridge, Git, AWS, AWS S3, LangGraph, AWS CDK, Github, Power BI, AWS EKS, langChain, Glue, Airflow, ECS, SonarQube, Python

Morgan Stanley: October 2020 – July 2023 (Location: New York, NY)

Role: Sr Data Engineer

Responsibilities:

- Designed and deployed production-grade data pipelines using **Python** and **Dagster** on **AWS**, ingesting PR metadata, commit metadata, and code quality telemetry from **GitHub** and **Jenkins**, integrating **AWS Lambda**, **AWS S3**, **AWS Glue**, and **AWS ECS** to ensure PCI-DSS-compliant financial workflows, automated retries, backfills, and observability with metrics, traces, and logs.
- Built retrieval agents using **Python** and **DSPy** to extract historical code context, diffs, and commit histories from **GitHub**, implementing prompt chaining and context packing for LLM optimization, automating score correction workflows, exposing CES and quality metrics via REST APIs, and ensuring SOX and GLBA-compliant secure audit trails across financial pipelines.
- Migrated orchestration from **Apache Airflow** to **Dagster**, enhancing PR/commit linking, branch tracking, and epic linking across **Jira**, **GitHub**, and banking systems, integrating **AWS EventBridge**, **AWS SQS**, and **AWS Lambda** to expose CES metrics and quality metrics to **Power BI** dashboards while maintaining PCI-DSS and SOX compliance.
- Developed anomaly detection and data quality monitoring using **Python**, **AWS Glue**, and **AWS CloudWatch**, identifying deviations in commit metadata, PR metadata, and code quality signals from **SonarQube** and **PMD**, triggering automated alerts, implementing DSPy-based score correction, and maintaining GLBA-compliant secure audit workflows for financial code review processes.
- Engineered AWS-native pipelines using **AWS ECS**, **AWS Lambda**, **AWS S3**, **AWS RDS**, and **AWS CDK** to automate backfills, retries, and scheduling for high-volume financial data ingestion, integrating **GitHub** webhooks and **Jenkins** CI/CD pipelines, storing datasets securely in **AWS S3** and **AWS RDS** with PCI-DSS encryption and IAM access controls.
- Implemented observability and monitoring frameworks using **AWS CloudWatch**, **AWS X-Ray**, and **Python** scripts, tracking metrics, traces, and logs across pipelines, triggering automated notifications through **AWS SQS** and Slack, ensuring PCI-DSS, SOX, and GLBA compliance while maintaining anomaly detection and continuous reliability of financial data workflows.
- Built backend services using **Python**, **AWS Lambda**, and **AWS EKS** to ingest PR metadata, commit metadata, and code quality signals from **GitHub**, **Jenkins**, **SonarQube**, and **PMD**, computing CES and quality metrics, exposing results via REST APIs with JWT authentication and **TLS 1.2+**, supporting PCI-DSS and SOX compliance.
- Designed and executed DSPy-based eval sets and feedback loops, collecting user feedback on scoring accuracy, retraining LLM/prompt optimization models stored in **AWS S3** and **AWS RDS**, implementing batching and caching to optimize performance and cost while ensuring GLBA and SOX-compliant secure financial code review workflows.
- Orchestrated data modeling and analytics workflows using **Python**, **AWS Glue**, **ERWIN**, and **E/R Studio** to design normalized schemas for PR/commit linking, branch tracking, and epic linking across **Jira**, **GitHub**, and banking systems, enabling quality metrics computation, engineering effort analysis, and PCI-DSS-compliant audit-ready reporting for executives via **Power BI**.
- Integrated **SonarQube** and **PMD** results with **Python** backend services to compute CES and quality metrics, linking findings to PR metadata, commit metadata, and **Jira** epics, deploying **AWS Lambda** functions triggered by **AWS EventBridge**, storing results in **AWS S3** and **AWS RDS**, ensuring PCI-DSS, SOX, and GLBA compliance.

- Developed retrieval agent workflows using **Python** and **DSPy** to pull code context, diffs, and commit history from **GitHub**, integrating pipelines with **Dagster** orchestration, enabling backfills and retries for historical data, exposing CES and quality metrics via JWT-secured APIs, ensuring SOX and GLBA-compliant secure audit logging.
- Built orchestration and reliability frameworks using **Dagster** and **AWS ECS**, implementing **AWS Lambda** and **AWS EventBridge** for event-driven ingestion of PR and commit metadata from **GitHub** and **Jenkins**, linking artifacts to **Jira** epics, enabling anomaly detection, data quality alerts, and PCI-DSS, SOX, and GLBA-compliant banking workflows.
- Engineered LLM/prompt optimization pipelines using **DSPy** and **Python**, creating eval sets from historical PR and commit metadata, automating score corrections via DSPy prompt tuning, optimizing LLM call performance through batching, caching, and prompt compression, ensuring secure, auditable, and compliant workflows aligned with GLBA and SOX standards.
- Designed data quality and observability frameworks using **Python**, **AWS CloudWatch**, **AWS X-Ray**, and **Dagster**, monitoring metrics, traces, and logs, integrating backend services with **SonarQube**, **PMD**, and **GitHub**, generating enriched datasets for **Power BI** dashboards, maintaining PCI-DSS, SOX, and GLBA compliance with secure storage in **AWS S3** and **AWS RDS**.
- Automated artifact linking using **Python** and **AWS Lambda**, connecting **Jira**, **GitHub**, PR metadata, and commit metadata for epic linking, work-type classification, and traceability, orchestrating pipelines in **Dagster**, storing datasets in **AWS S3** and **AWS RDS**, implementing anomaly detection, and ensuring PCI-DSS, SOX, and GLBA-compliant financial code review processes.
- Designed and deployed PCI-DSS-compliant data ingestion frameworks using **AWS EKS**, orchestrating containerized microservices with **Airflow** for ETL workflows that processed sensitive cardholder data, implementing end-to-end encryption, audit logging, RBAC, and autoscaling policies to enhance reliability and security for high-volume banking operations.
- Implemented **AWS WAF** rules and security policies for **AWS EC2**-hosted banking applications, mitigating SQL injection, XSS, and DDoS attacks, integrating CloudWatch monitoring, geo-blocking, and rate-limiting configurations, ensuring GLBA compliance, secure customer authentication workflows, and reduced exposure to vulnerabilities across digital banking platforms.
- Developed infrastructure-as-code templates using **AWS CDK** in TypeScript to provision **AWS EC2**, VPCs, security groups, IAM roles, and backup policies for loan origination systems, automating deployments, enforcing least-privilege access, and ensuring PCI-DSS and SOX compliance with version-controlled, repeatable provisioning pipelines.
- Maintained version control and collaborative workflows using **Git** and **ERWIN**, managing database schema changes, branch strategies, CI/CD integration, and schema validation, ensuring audit trail consistency, peer-reviewed modifications, and GLBA-compliant governance across customer account, transaction, and mortgage data models in production banking environments.
- Configured **Airflow** DAGs to automate inter-bank settlement reconciliations, orchestrating validation, anomaly detection, and report generation processes, enhancing task retries and dependencies, ensuring timely delivery of regulatory reports, SOX compliance, and SLA adherence, improving reliability for financial data processing and internal audit requirements.
- Collaborated with data architects using **E/R Studio** to reverse-engineer banking databases, producing entity-relationship diagrams and data dictionaries, identifying PII and sensitive data elements, assessing downstream dependencies, and supporting GLBA and PCI-DSS protections during schema migrations and core banking modernization initiatives.
- Provisioned and managed **AWS EC2** instances for batch processing of interest calculations, fee assessments, and statement generation, implementing multi-AZ deployments, automated snapshots, CloudWatch alarms, auto-recovery, and capacity planning to ensure high availability, disaster recovery readiness, and regulatory compliance for retail banking operations.
- Integrated **ERWIN** metadata repositories with **Git**, centralizing logical and physical data models across mortgage, credit card, and commercial lending domains, automating model exports to Git, enforcing naming conventions, version control, and change management to ensure GLBA-compliant governance and auditable production deployments in regulated financial services environments.

Environments: Lambda, Jenkins, Airflow, S3, AWS Lambda, SonarQube, Dagster, ERWIN, JWT, E/R studio, ECS, AWS EC2, AWS RDS, Power BI, AWS EKS, AWS WAF, Jira, AWS CDK, DSPy, Python, Glue, AWS S3, Git, AWS, PMD, Github, EventBridge, AWS SQS

Delta Airlines: December 2017 – September 2020 (Location: Atlanta, GA)

Role: Data Engineer / Data Analyst

Responsibilities:

- Designed and deployed production-grade data pipelines using **Python** and **Apache Airflow** to ingest commit metadata and PR metadata from **GitHub**, integrating **AWS Lambda** and **AWS EventBridge** for real-time processing, exposing quality metrics via REST APIs, ensuring SOC 2 audit trails and FAA/IATA compliance across aviation engineering workflows.
- Built end-to-end data modeling frameworks using **Python** to link PRs and commits, tracking branch merges to main repositories, integrating **AWS Glue** and **Dagster** pipelines for historical backfills, implementing **Prometheus** and **Grafana** dashboards for anomaly detection and IOSA-compliant aviation software quality monitoring.
- Engineered retrieval agents using **Python** to extract contextual metadata from **GitHub**, integrating **SonarQube** outputs into **AWS S3** data lakes, computing technical debt and maintainability metrics, and exposing metrics via **Power BI**, ensuring observability through **Prometheus** and logs for continuous FAA-compliant quality improvement.
- Orchestrated migration of legacy workflows using **Dagster** and **Airflow** on **AWS ECS** and **AWS EKS**, implementing retry logic, scheduling policies, and monitoring frameworks, triggering **AWS Lambda** via **AWS EventBridge**, integrating **Grafana** and **AWS SNS** notifications to maintain 99.9% pipeline uptime aligned with SOC 2 and IATA standards.
- Developed analytics platforms using **Python**, **AWS RDS**, and **Power BI** to generate CES metrics, correlating PR and commit metadata with **Jira** epics, designing schemas in **ERWIN** and **E/R Studio** to support graph-like linking, enabling epic-level reporting aligned with FAA documentation and safety-critical software audit readiness.
- Implemented backend services using **Python** on **AWS Lambda** and **AWS EC2**, processing thousands of daily PRs and commits, integrating **Jenkins** CI/CD pipelines with **PMD** static analysis, storing data in **AWS S3**, ensuring IOSA-compliant traceability, FAA version control governance, and safety-critical code quality monitoring.
- Built observability frameworks using **Prometheus**, **Grafana**, and **AWS CloudWatch**, tracking Airflow DAG execution times, **AWS Glue** job success rates, and Lambda invocation patterns, implementing Python scripts and S3 pipelines for audit trail logging, enabling SOC 2 compliance and root-cause analysis of aviation software pipelines.
- Orchestrated backfills of historical **GitHub** repository data using **Dagster** and **AWS Glue**, reconstructing branch/main tracking histories, implementing Python-based incremental processing, defining Prometheus-based data quality alerts, detecting duplicate PRs, timestamp inconsistencies, and ensuring FAA-compliant anomaly detection for safety-critical aviation releases.
- Developed eval set creation workflows using **Python**, integrating **Jira** and **GitHub** APIs for feedback-driven quality validation, implementing **AWS ElastiCache** caching and Lambda batching optimizations, improving pipeline efficiency while maintaining observability in **Grafana**, supporting IATA-compliant quality scoring and engineering metrics.
- Designed normalized data schemas for PR/commit linking and epic metadata using **ERWIN** and **E/R Studio**, deploying Python services on **AWS ECS**, storing data in **AWS RDS** and **AWS S3**, maintaining FAA-compliant audit trails, supporting aircraft program-specific quality metrics and work-type classification for aviation engineering teams.
- Established observability standards using **Prometheus**, **Grafana**, and **AWS CloudWatch**, implementing Python scripts to detect anomalies in commit ingestion, backfill failures, or pipeline delays, configuring **AWS SNS** notifications to ensure rapid alerting, maintaining operational reliability standards for IATA-compliant flight operations and maintenance systems.
- Integrated **SonarQube** with **Airflow** and **AWS Glue**, ingesting static analysis metrics into **AWS S3**, mapping code severity to FAA DAL levels, implementing Python logic to generate automated risk scoring, and visualizing defect densities, technical debt, and maintainability indices via **Power BI** for aviation software compliance.
- Designed backend services using **Python**, **AWS Lambda**, **AWS SQS**, and **AWS EventBridge**, implementing asynchronous PR/commit processing, configuring **AWS CDK** to provision **AWS ECS**, **AWS RDS**, and **AWS S3**, enforcing AWS WAF security policies, and integrating Prometheus alerts for ingestion lag, anomaly detection, and aviation quality monitoring.
- Built analytics dashboards in **Power BI** connected to **AWS RDS** and **AWS S3**, computing trends in code review turnaround, defect escape rates, and test coverage, aggregating data using **Python** and **AWS Glue**, maintaining FAA-compliant documentation, integrating **Collibra** for data catalog governance and audit readiness for safety-critical systems.
- Established governance frameworks using **Collibra**, documenting PR/commit ingestion workflows, data modeling schemas, and quality metric calculations, automating Python scripts integrated with **Jenkins** CI/CD pipelines to enforce anomaly detection thresholds, trigger **AWS SNS** alerts, and maintain SOC 2-compliant aviation software engineering metrics.
- Developed flight operations platforms using **DSPy**, integrating **AWS SQS** for asynchronous messaging between microservices handling flight updates, maintenance alerts, and crew notifications, implementing batching strategies for throughput optimization, ensuring FAA Part 121 operational integrity and IATA-compliant timely dissemination of safety-critical information.
- Built scalable infrastructure-as-code solutions using **AWS CDK** to provision **AWS EKS** clusters for aviation safety applications, integrating Git version control, branch protection strategies, and **AWS WAF** with custom rules to safeguard flight planning APIs, ensuring SOC 2 Type II and IOSA compliance for sensitive operational data.
- Implemented distributed caching architecture using **AWS ElastiCache**, optimizing frequently accessed aircraft maintenance records, integrating **PMD** static analysis in CI/CD pipelines for Java-based aviation systems, ensuring FAA

maintenance approval compliance, and improving response times for pre-flight inspections while maintaining safety-critical coding standards.

- Configured **AWS EC2** Auto Scaling groups to run flight analysis workloads, leveraging **DSPy** NLP frameworks to process pilot and maintenance reports, deploying multi-AZ instances for failover, supporting FAA SWIM real-time data exchange requirements, and maintaining IATA-compliant operational tracking across flight and baggage systems.

Environments: AWS CDK, Python, ERWIN, AWS EC2, EventBridge, github, SonarQube, PMD, AWS Lambda, Prometheus, AWS EKS, Dagster, Airflow, AWS, DSPy, AWS WAF, AWS S3, ECS, Glue, Jenkins, Lambda, AWS RDS, AWS Glue, S3, E/R studio, Grafana, Power BI, Git, AWS SQS, Colibra, AWS SNS, Jira

Cisco: February 2016 – November 2017 (Location: San Jose, CA)

Role: Data Engineer / Migration Specialist

Responsibilities:

- Built a Network Data Integration platform using **Azure Cloud**, improving enterprise data integration efficiency while enabling seamless data flow across global networking systems by consolidating monitoring telemetry from thousands of devices deployed in international data centers.
- Developed scalable data ingestion pipelines using **Talend ETL** and **Azure Data Factory**, efficiently processing large volumes of network monitoring and security appliance data while ensuring reliable ingestion, transformation, and availability for downstream operational analytics.
- Executed data migration initiatives transferring configuration data from legacy network management systems to modern **SDN platforms**, maintaining configuration accuracy and uninterrupted network services throughout the transition to software defined infrastructure.
- Built complex orchestration workflows using **Apache Airflow**, coordinating ETL processes across distributed network monitoring systems in multiple global regions while aligning execution schedules with regional compliance and operational time zone requirements.
- Implemented **Azure SQL Database** to store structured network performance metrics, optimizing data models and query execution to support rapid access to critical network health indicators for operations and engineering teams.
- Configured **Azure Data Lake Storage** for scalable retention of semi structured and unstructured network traffic data, supporting long term security analysis, capacity planning, and cost efficient storage of historical traffic patterns.
- Integrated diverse network data sources including router logs, switch telemetry, and security appliance alerts into unified monitoring solutions, enabling cross layer correlation and faster troubleshooting across complex global network environments.
- Implemented data security and privacy controls aligned with international regulations, applying **encryption**, **role based access controls**, and **data anonymization** techniques while handling sensitive network configuration and traffic analysis datasets.
- Executed migration of legacy network management platforms to modern cloud based monitoring solutions, ensuring adherence to industry standards while improving observability and reducing operational overhead for network operations teams.
- Developed predictive capacity forecasting models using **Azure Databricks**, analyzing historical traffic trends to improve bandwidth planning accuracy and support proactive decision making for global network growth requirements.
- Created real time network anomaly detection solutions using **streaming analytics**, enabling early identification of security threats and performance degradation by processing high volume network event streams continuously.
- Designed and implemented data quality frameworks validating global network configuration data, ensuring consistency and correctness of configuration changes prior to deployment across distributed infrastructure environments.
- Built custom integration APIs enabling connectivity between network management platforms and **IT Service Management systems**, supporting automated incident creation, updates, and resolution workflows across operational teams.
- Developed interactive dashboards visualizing global network performance and security posture using **business intelligence tools**, providing leadership with real time visibility into service levels and infrastructure health.
- Implemented automated network capacity planning solutions leveraging historical traffic analysis and growth projections, optimizing equipment utilization and supporting informed infrastructure investment decisions across global operations.

Environments: Azure Cloud, Talend ETL, Azure Data Factory (ADF), Apache Airflow, Azure SQL Database, Azure Data Lake Storage, Azure Databricks, Python, Streaming Analytics, SDN Platforms

Flipkart: August 2014 – December 2015 (Location: Bangalore, India)

Role: ETL Developer

Responsibilities:

- Built a scalable ETL pipeline using **AWS Services**, **Informatica**, and **Talend**, integrating data from e commerce platforms, inventory systems, and customer databases to create a unified data flow supporting high volume daily transaction processing across multiple online retail channels.
- Configured **Amazon S3** for secure storage of raw, staging, and transformed product catalog datasets, enabling reliable ingestion workflows for large SKU volumes while applying versioning and lifecycle policies to balance accessibility with long term storage cost optimization.
- Utilized **Informatica PowerCenter** to extract data from **MySQL**, **PostgreSQL**, and **MongoDB** sources containing customer profiles and purchase history, ensuring accurate extraction, preserved relationships, and consistent processing of high volume customer records.
- Developed optimized **Talend** jobs processing **CSV** and **JSON** based vendor feeds and marketplace listings, improving data processing efficiency and supporting timely product catalog updates across web and mobile commerce platforms.
- Implemented complex data transformations using **Informatica**, performing data cleansing, validation, and standardization on product metadata to improve search accuracy and ensure consistent product taxonomy across multiple vendor sources.
- Configured **Talend Open Studio** to perform data integrations between order management and fulfillment systems, streamlining ETL workflows and synchronizing inventory levels across distributed warehouse and fulfillment center operations.
- Designed **AWS Glue** jobs to catalog, transform, and load curated sales and inventory data into **Amazon Redshift**, optimizing storage layouts and enabling efficient analytical queries supporting sales performance and category level insights.
- Automated ETL job scheduling using **AWS Lambda** and **Amazon CloudWatch Events**, improving reliability of time sensitive processes such as daily sales reporting and inventory refreshes through managed execution and dependency handling.
- Monitored ETL pipeline performance using **Amazon CloudWatch**, configuring alerts for failures and latency issues in critical data flows supporting order processing and financial reconciliation activities.
- Connected **Tableau** to **Amazon Redshift** to deliver interactive dashboards for real time analysis of sales trends, inventory positions, and customer behavior, enabling merchandising teams to make informed decisions on product placement and promotions strategies.

Environments: AWS (S3, Glue, Lambda, CloudWatch, Redshift), Informatica PowerCenter, Talend, Talend Open Studio, MySQL, PostgreSQL, MongoDB, CSV/JSON files, Tableau

EDUCATION:

- Bachelor of Technology in Computer Science & Engineering at Koneru Lakshmaiah University 2010-2014